European Journal of Medicinal Chemistry 148 (2018) 423–435

Contents lists available at ScienceDirect

European Journal of Medicinal Chemistry

journal homepage: http://www.elsevier.com/locate/ejmech

Research paper

Precise structures of fucosylated glycosaminoglycan and its oligosaccharides as novel intrinsic factor Xase inhibitors

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ARTICLE INFO

Article history:
Received 10 November 2017
Received in revised form
24 January 2018
Accepted 14 February 2018
Available online 15 February 2018

Keywords:
Glycosaminoglycan
Oligosaccharide
Anticoagulant
Intrinsic factor Xase

ABSTRACT

Selective inhibition of the endogenous coagulation pathway is a promising strategy for developing new anticoagulants. Fucosylated glycosaminoglycan (FG), a structurally complex glycosaminoglycan, has distinct anticoagulant properties, especially the strong intrinsic factor Xase inhibitory activity that is recognized as a new target with potential physiological and therapeutic applications. Detailed knowledge of FG structures is necessary for developing a clinically effective intrinsic FXase inhibitor. However, challenges remain to elucidate FG structures as a basis for pharmaceutical development. Herein, using the highly selective β-elimination method, oligosaccharides with regular structures were prepared from the depolymerization products. Analysis of oligosaccharides further confirmed the precise structural sequence of the FG. Furthermore, biological activity assay suggested that these pure homogeneous oligosaccharides, particularly an octasaccharide, exhibit strong inhibition of the intrinsic coagulation pathway by inhibiting human intrinsic factor Xase. Our finding is significant for discovery of a new class of anticoagulant agents as intrinsic factor Xase inhibitors.

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1. Introduction

Fucosylated glycosaminoglycan (FG) is an unusual glycosaminoglycan (GAG) found, so far, exclusively in sea cucumbers (Echinodermata, Holothuroidea). It possesses a chondroitin sulfate-like backbone and is markedly different from typical mammalian GAGs because of its unique sulfated fucose side chains [1]. FG has recently attracted considerable attention because of its various bioactivities related to inflammation, angiogenesis, cellular growth, cancer metastasis, fibrosis, virus infection, hyperglycemia, atherosclerosis, and, above all, coagulation and thrombosis [1]. Interestingly, its anticoagulant and antithrombotic activity seems to be achieved by serpin independently inhibiting the intrinsic factor Xase complex (FXase, factor IXa-VIIIa-Ca^{2+}-PL complex) [2,3]. Notably, inhibiting components in the intrinsic coagulation pathway, such as factor XIIa, XIa, IXa and FXase complex, should effectively prevent thrombus formation, with negligible bleeding risks [4,5]. Intrinsic FXase complex seems to be an ideal target for new antithrombotic agents, but no intrinsic FXase inhibitors have yet been reported, except for the some oligosaccharides that we previously described [6]. In particular, by elucidating the relationship between structures and anticoagulant activities of FG, a low molecular weight FG has very recently been suggested as a promising candidate for the development of an antithrombotic drug that would be safer and more effective than low-molecular-weight heparin [7–9]. Consequently, detailed knowledge of the native FG structure or its fragments is also necessary for developing a clinically effective inhibitor of intrinsic FXase complex that has less side effects.

As a biomacromolecule, FG has very complex structure, and those from various sea cucumber species are significantly different [1]. Disparate FG structures can even be isolated from the same species, although possibly due to different methods used in the

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https://doi.org/10.1016/j.ejmech.2018.02.047
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process of purification and analysis [10–14]. The detailed structures of FG have not been elucidated because this polysaccharide is heterogeneous—it is a mixture of isomers with different molecular weights, and available techniques are still limited for analyzing such molecules. Therefore, describing the precise structures of FGs remains a challenge for progressing the development of pharmaceuticals.

For over thirty years since its discovery, some chemical research teams, including our group, have made advances in clarifying FG structures. Research studies have shown that FG is a kind of glycosaminoglycan with a basic structure comprising a chondroitin sulfate backbone and sulfated fucoses as the side chains [10–22]. However, there have been differing conclusions about the linkage sites of sulfated fucoses and substitution patterns of sulfate esters. For example, although it is assumed that a single fucose (Fuc) is linked to the *C*-3 position of glucuronic acid (GlcA) *via* an α-glycosidic bond, there is no direct evidence to exclude the possibilities that Fuc may be linked to the *C*-4 and *C*-6 positions of *N*-acetylgalactosamine (GalNAc) and that one Fuc side chain may comprise a di- or trisaccharide [10–22].

Directly analyzing the polysaccharides as a macromolecule remains difficult using current techniques such as high resolution nuclear magnetic resonance (NMR) and mass spectrometry (MS) [23,24]. By comparison, depolymerization is an effective method for elucidating the precise structure of polysaccharides [16,25–27]. Since enzymes that can be used for enzymatic hydrolysis of FG have not been discovered, chemical depolymerization methods have been established, such as free radical depolymerization and the ^{60}Co irradiation method [16,25–27]. But providing direct evidence of chemical structures by these methods is still difficult, owing to the lack of selectivity of glycosidic bonds, which results in difficulty in purifying or/and analyzing the oligosaccharide products.

In this work, we sought to develop a selective degradation method like enzymatic hydrolysis, which is favorable to obtain the products with certain structural characteristics. We further undertook to purify and analyze the oligosaccharides from these low-molecular-weight products, in order to obtain the structure information and finally elucidate the structure of the native FGs. Based on this strategy, we prepared a new class of homogeneous oligosaccharides of FG using our newly developed selective β-eliminative method. Analysis of these oligosaccharides revealed the precise structure of FG. Furthermore, anticoagulant evaluation of these oligosaccharides revealed the structure–activity relationship (SAR), providing a pathway for the discovery of novel anticoagulants.

2. Results and discussion

2.1. Native FG compounds

Recently, our group had extracted and purified some abundant representative FGs from dozens of various sea cucumber species by the methods described previously [7,28,29]. The chemical structures of these FGs were analyzed by studying their monosaccharide compositions, functional groups, and NMR spectra. The results indicated that they share certain structural features, i.e. a chondroitin sulfate (CS)-like backbone and side chains whose sulfated fucose units linked to the *C*-3 position of β-*D*-glucuronic acids. Further, the sulfation patterns of L-fucose (Fuc) residues varied from one to another, as identified by 1D and 2D NMR spectra [7,28,29].

Among the FGs from these sea cucumbers, FG **1**, a possible regular FG with molecular weight of ~46 kDa from the tropical sea cucumber *Stichopus horrens*, was selected for further study, because of its favorable characteristics for elucidating the detailed structure of FG. Our results showed that FG **1** contains about 92% 2, 4-di-*O*-sulfated fucose (Fuc_{2S4S}) and a small amount of other types of sulfated fucoses (Fig. S1, see the Supporting Information). Naturally, our strategy to research **1** can also be applied to other FGs from various species of sea cucumber.

As shown in the NMR spectrum of **1** (Fig. S1), signals of monosaccharide residues could not be assigned, for they gave overlapping spectra with broad signals, thus hindering the elucidation of precise structures [7]. In particular, the residues cause difficulty in identifying the connection patterns of saccharides in the backbone and the Fuc side chains.

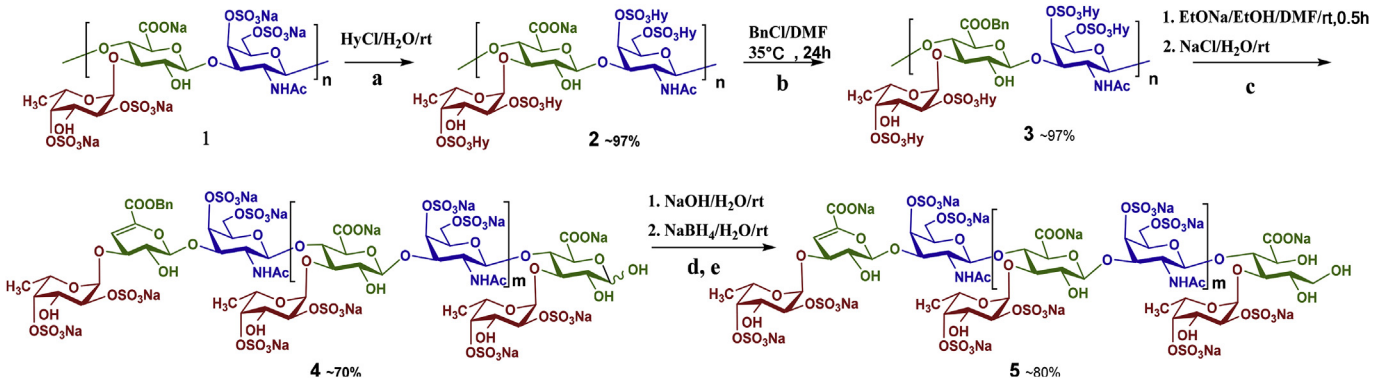
2.2. Preparation of pure oligosaccharides of FG by β-eliminative depolymerization

Enzymatic cleavage of GAGs is an effective method to prepare oligosaccharides with unsaturated uronic acid at the non-reducing end, which can be applied to analysis of their chemical structures [30]. But unlike linear GAGs, FGs cannot be directly digested by known polysaccharide hydrolyase such as heparanase or chondroitin lyase, owing to the steric hindrance caused by their Fuc side chains [10–25,28,29]. Fortunately, the component of glucuronic acid should allow FGs to be β-eliminated in order to remove the substituent attached at *O*-4. This is because β-elimination is considered to be a selective method for depolymerization of polysaccharides having uronic acid by breaking the 1 → 4 glycosidic linkage and forming $\Delta^{4,5}$ unsaturated uronic acid residues at the non-reducing end of resulting fragments [31–34]. In the present study, we further developed a selective depolymerization and found novel results.

Sodium salt of FG **1** was firstly transformed to benzethonium salt **2** (Scheme 1a). Next, product **2** was producted to benzyl ester **3** by esterification with benzyl chloride in DMF (Scheme 1b). Product **3** was then treated with freshly prepared 0.02 M EtONa/EtOH at room temperature for about 30 min and transformed to product **4**, a sodium salt of benzyl substituted depolymerized product (Scheme 1c). The benzyl substituents in **4** were removed by saponification (Scheme 1d) and its reducing ends were reduced to alcoholic hydroxyl using $NaBH_4$ (Scheme 1e). Lastly, the end product dFG **5** were obtained with a yield of ~55% based on the starting material **1** and the structure was analyzed.

For heparin, β-eliminative depolymerization can take place in alkaline aqueous solution [35], while it was not observed on FG carboxylic ester in the same solution [34]. When the carboxylic ester of FG was treated with alkaline in a non-aqueous solvent, the reaction solution showed strong UV absorption at 234 nm, indicating that the FG ester could be cleaved by β-elimination and distinctive unsaturated bonds could be formed (Fig. 1A, Fig. S2). This characteristic could be applied for chromatography analysis or purification of these oligosaccharides obtained by depolymerization (Fig. 1).

Apart from FGs, the sea cucumbers also usually contain other types of polysaccharides, such as neutral glycan or/and fucan sulfate (FS) that do not contain uronic acid [29,36–38]. Due to the similarity of molecular weight and charge density between them, it is very difficult to isolate FG by physical separation methods such as gel exclusion and ion exchange chromatography. This might also be an important reason underlying the FG structural controversy. However, by the β-eliminative depolymerization method that we established, the glycosidic bonds linked to uronic acid can be selectively cleaved, thus the interference of other polysaccharides can be avoided. Indeed, when the depolymerization of a purified natural FG from *S. horrens* was carried out by this new method, another polysaccharide, could not be degraded, confirming to the high selectivity of our method to FG. This polysaccharide is a fucan sulfate, possessing similar molecular weight and different



Scheme 1. Preparation of FG-derived oligosaccharides. Sodium salt of FG **1** was firstly transformed to benzethonium salt **2** (a); next, product **2** was transformed to its benzyl ester by esterification with benzyl chloride in DMF (b); product **3** was depolymerized by EtONa and transformed to sodium salt **4** (c); the benzyl substituents in **4** were removed by saponification (d) and its reducing ends were reduced to alcoholic hydroxyl using $NaBH_4$ (e), then the end product dFG **5** was obtained and the structure was analyzed. Detailed reaction conditions are described in *Experimental Section*.

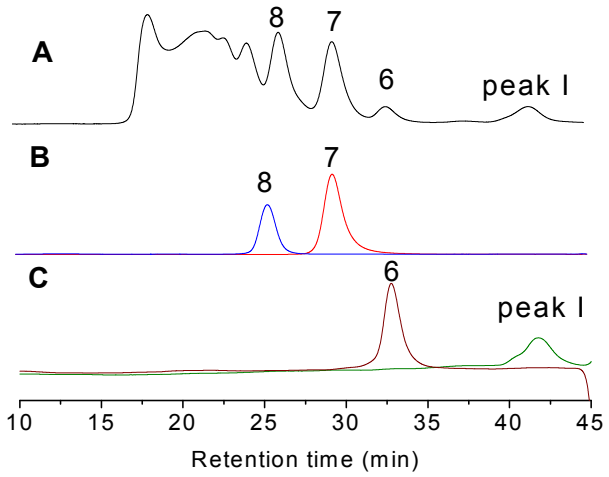


Fig. 1. HPGPC profiles of dFG (**5**) (A), compounds **7**~**8** (B) and compound **6** and peak I (C). HPGPC profiles were analyzed using a Superdex Peptide 10/300 GL column (10 mm × 300 mm) at 234 nm or RID. Aliquots of 50 μL of 2 mg/mL samples were analyzed on an Agilent Technologies 1200 series eluted with 0.2 M NaCl solution at a flow rate of 0.4 mL/min.

monosaccharide composition (Fig. S3). Its structural analysis is currently underway, and the results will be published in due course.

The depolymerized product dFG 5 was characterized by $\Delta^{4,5}$ unsaturated glucuronic acid at the non-reducing end, as confirmed by the 1D/2D NMR spectra assignments (Fig. 2, Fig. S4). In the ^{1}H NMR spectrum of **5**, a group of distinct signals at 5.74 ppm, which was absent in that of the native FG, could be assigned to the proton *H*-4 of $\Delta^{4,5}$-unsaturated glucuronic acids (ΔU). This was in good agreement with the corresponding signals of heparin and chondroitin sulfate fragments produced by β-eliminative degradation [32]. In the ^{13}C NMR spectrum, the most distinct signal at 149.9 ppm was observed, and could be assigned to olefinic *C*-5 of ΔU given the presence of a carboxyl group. Another distinct signal at 109.2 ppm was assigned to *C*-4 of ΔU, which was found to be correlated with *H*-4 of ΔU as shown in the HSQC spectrum. The *C*-1, *C*-2 and *C*-3 signals from ΔU appeared approximately at 106.0 ppm, 73.2 ppm and 79.1 ppm, respectively. These data indicated that the GalNAc (1 → 4) GlcA glycosidic bonds were selectively cleaved, producing distinctive $\Delta^{4,5}$ unsaturated residues at the non-reducing end.

Although the NMR spectra of **5** are clearer than those of **1**, the signals of product **5** overlapped markedly due to its large molecular size and composition of isomers. Thus, we could only roughly resolve the NMR spectrum data (Fig. S4, Table S1) based on our results of a more complex FG from the sea cucumber *Thelenota ananas* [34]. Additionally, the structural information of dFG deduced from NMR data and chemical characteristics may be insufficient to provide direct evidence for the native FG structure. Because the dFG is not homogenous, being composed of oligosaccharides with special non-reducing ends and reducing termini, the NMR signals can be complicated. Therefore, we sought to clarify the structure of **1** by analyzing purified oligosaccharides in dFG 5 using a bottom-up approach.

Firstly, the depolymerized product **5**, which is a mixture of fragments (M_W 7.2 kDa), was detected by high-performance gel permeation chromatography (GPC) (Fig. 1A, Fig. S2C). Next, the oligosaccharides were purified from dFG **5** by GPC using Bio-Gel P-2, P-6 and P-10 columns. Each peak was collected according to the elution curve at 234 nm, analyzed by a Superdex Peptide 10/300 GL column and then desalted on a column packed with Sephadex G-10 (detailed protocol shown in *Experimental Section* 4.4). Finally, a series of homogeneous oligosaccharides were obtained (Fig. 1) and identified as tri-, penta-, and octa-saccharide, respectively, according to ^{1}H NMR spectra. It displayed the peak area ratio of the anomeric proton of Fuc residues at different positions of the chains (Figs. S5–7). The desalted, pure oligosaccharides were water-soluble white powders (HPLC purity > 98%) (compounds **6**~**8**).

2.3. Structural characterization of oligosaccharides

The NMR spectra clearly indicated that compound **7** is a pentasaccharide based on the ratio of the *H*-1 proton integrals of each sugar in ^{1}H NMR (Fig. S5A) and the number of spin systems in ^{1}H-^{1}H COSY (Fig. S5C). The complete assignment of its signal peaks is given in Table S2. The five *H*-1 proton signals at 4.928, 5.516, 4.805, 3.813 and 5.235 ppm were assigned to $\Delta^{4,5}$-unsaturated glucuronic acid (ΔU), sulfated fucose linked to ΔU (dF), *N*-acetyl-2-deoxy-2-amino-galactose (GalNAc, A), the corresponding alcohol of glucuronic acid (GlcA-ol, rU) produced by sodium borohydride and the sulfated fucose (rF) linked to this GlcA-ol, respectively. Their *C*-1 chemical shifts were at 105.94, 99.30, 103.63, 65.22 and 101.38 ppm, respectively (Fig. S5B). Resonance of rU-1 at 65.22 ppm was the $-CH_2$ which is a reducing end of GlcA-ol unit. The locations of the attached sulfates on each sugar residue were deduced from the down-field chemical shifts of protons on attached carbons

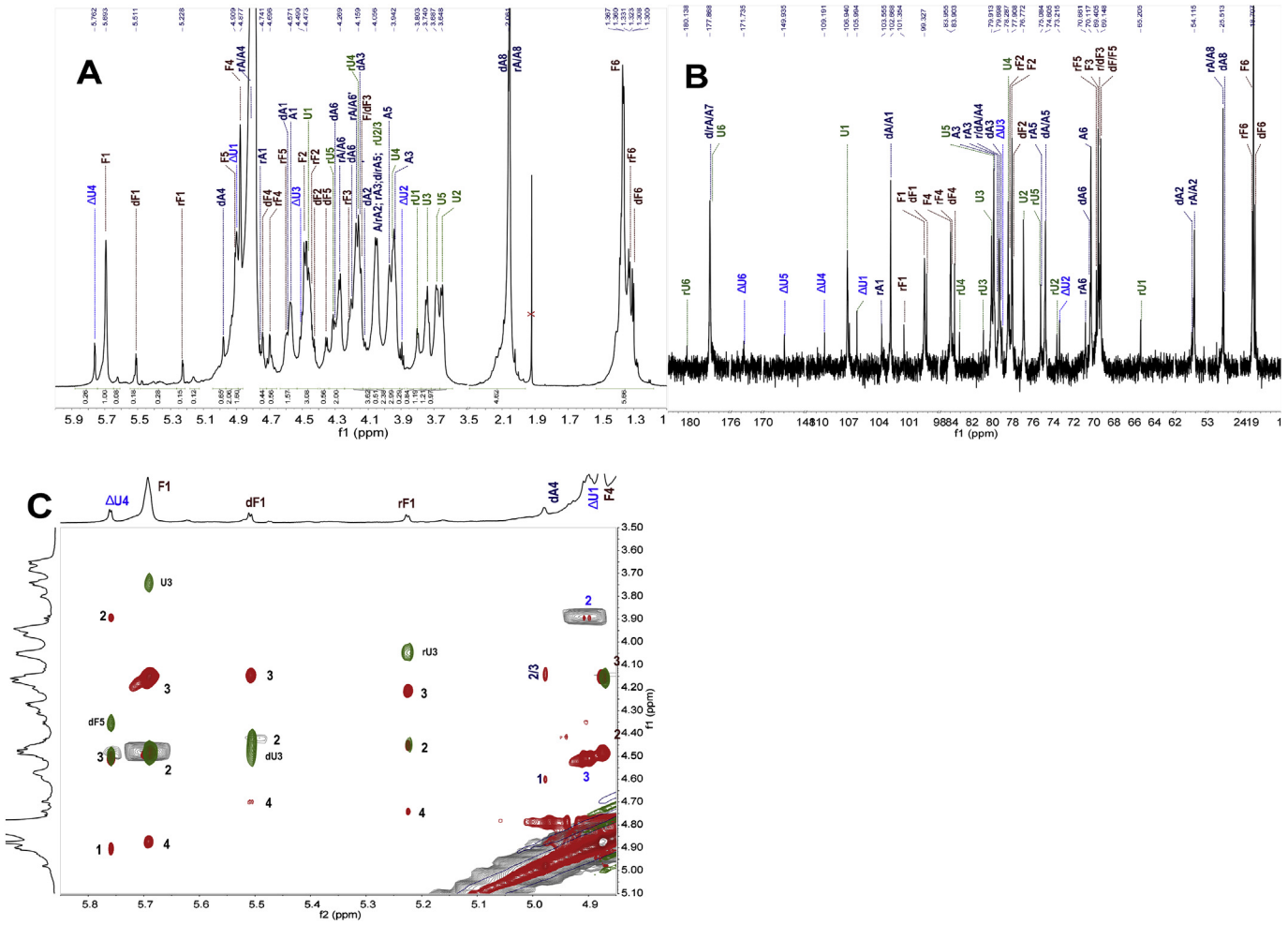


Fig. 2. ^{1}H (A), ^{13}C (B) and ^{1}H-^{1}H 2D (C) NMR spectra of depolymerized FG **5**. (COSY, black; TOCSY, red; ROESY, green). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

caused by the sulfation compared with the corresponding unsubstituted monosaccharide. This indicated that GalNAc is sulfated at both the *C*-4 and *C*-6 positions and that Fuc is sulfated at both the *C*-2 and *C*-4 positions. Additionally, the HSQC spectrum (Fig. 3B) confirmed these sulfated positions on Fuc and GalNAc. The small *H*-*H* coupling constant (3.84 Hz) indicated the linkage between Fuc and ΔU was α-glycosidic bond, and the larger *H*-*H* coupling constant (8.48 Hz) indicated the presence of a β linkage between ΔU and GalNAc. The intraresidue and interresidue correlation signals were observed in the ROESY spectrum (Fig. 3A). It indicated that the Fuc is connected to the *C*-3 position of ΔU and that β(1 → 3) connection occurs between ΔU and GalNAc, and GalNAc is linked to *C*-4 of GlcA-ol. These glycosidic linkages of compound **7** were confirmed by further analysis using HMBC spectrum (Fig. S5E). Additionally, the ESI-Q-TOF-MS of **7** afforded a molecular weight value of 1504.8664 Da. The result is accordant with that of $[M - Na]^-$ (observed $m/z = 1481.8766$), $[M - 2Na]^{2-}$ (observed $m/z = 729.4430$) (Table S3), indicating a molecular formula of $C_{32}H_{43}O_{54}NS_6Na_8$ (Fig. 3C). Further assignments (Table S2) determined compound **7** as a pentasaccharide with the structure of [*L*-Fuc$_{2S4S}$]-α(1 → 3)-*D*-ΔU-β(1 → 3)-*D*-GalNAc$_{4S6S}$-β(1 → 4)-[*L*-Fuc$_{2S4S}$]-α(1 → 3)-*D*-GlcA-ol.

Compound **8** showed chemical signals more complex than those from **7** according to 1D and 2D NMR analyses (Fig. 4, Fig. S6). Besides ΔU, dF, rF and rU, the ^{1}H NMR spectrum of **8** showed additional *H*-1 signal peaks of Fuc$_{2S4S}$ (F) at 5.700 ppm, *H*-1 signal of GlcA (U) at 4.477 ppm and GalNAc linked to ΔU (dA) or GlcA-ol (rA), which could be easily assigned according to COSY. Similarly, compared with that of **7**, the ^{13}C NMR spectrum of **8** showed additional terminal C signals, which could be further assigned by HSQC (Fig. S6E). All ^{1}H and ^{13}C NMR signals (Fig. 3) could be well assigned by 2D NMR (TOCSY and HSQC) (Figs. S6D–E). The *H*-5 of the additional fucose residue in the internal repeating units showed the anomalously high chemical shift at 4.939 ppm caused by the formation of a hydrogen bond between *C*-*H* and the oxygen ring (*C*-*H*···*O*) of the preceding GalNAc, in accordance with that of native FG [39]. However, the fucose linked to ΔU at the non-reducing end (dF) in compound **8** did not have this arrangement, hence its *H*-5 chemical shift returned the expected value of 4.353 ppm. These results suggest that an additional trisaccharide may exist in **1** as a repeating unit. The ROESY and HMBC spectra (Fig. 4A, Figure 6B) showed that the glycosidic linkages of the additional trisaccharide unit and the glycosidic linkages were consistent with that of compound **7**. The α or β linkages could also be determined by the *H*-*H* or *C*-*H* coupling constants. **8** had the characteristic mass-to-charge peaks of $[M - 2Na]^{2-}$ (observed m/z 1206.8710, theoretical m/z 1206.8991), $[M - 3Na]^{3-}$ (observed m/z 796.9158, theoretical m/z 796.9361), and others in its ESI-Q-TOF-MS spectrum (Fig. 4C). ESI-Q-TOF-MS and NMR analysis suggests that **8** is an octasaccharide and possesses the structure of [*L*-Fuc$_{2S4S}$-α(1 → 3)]-*D*-

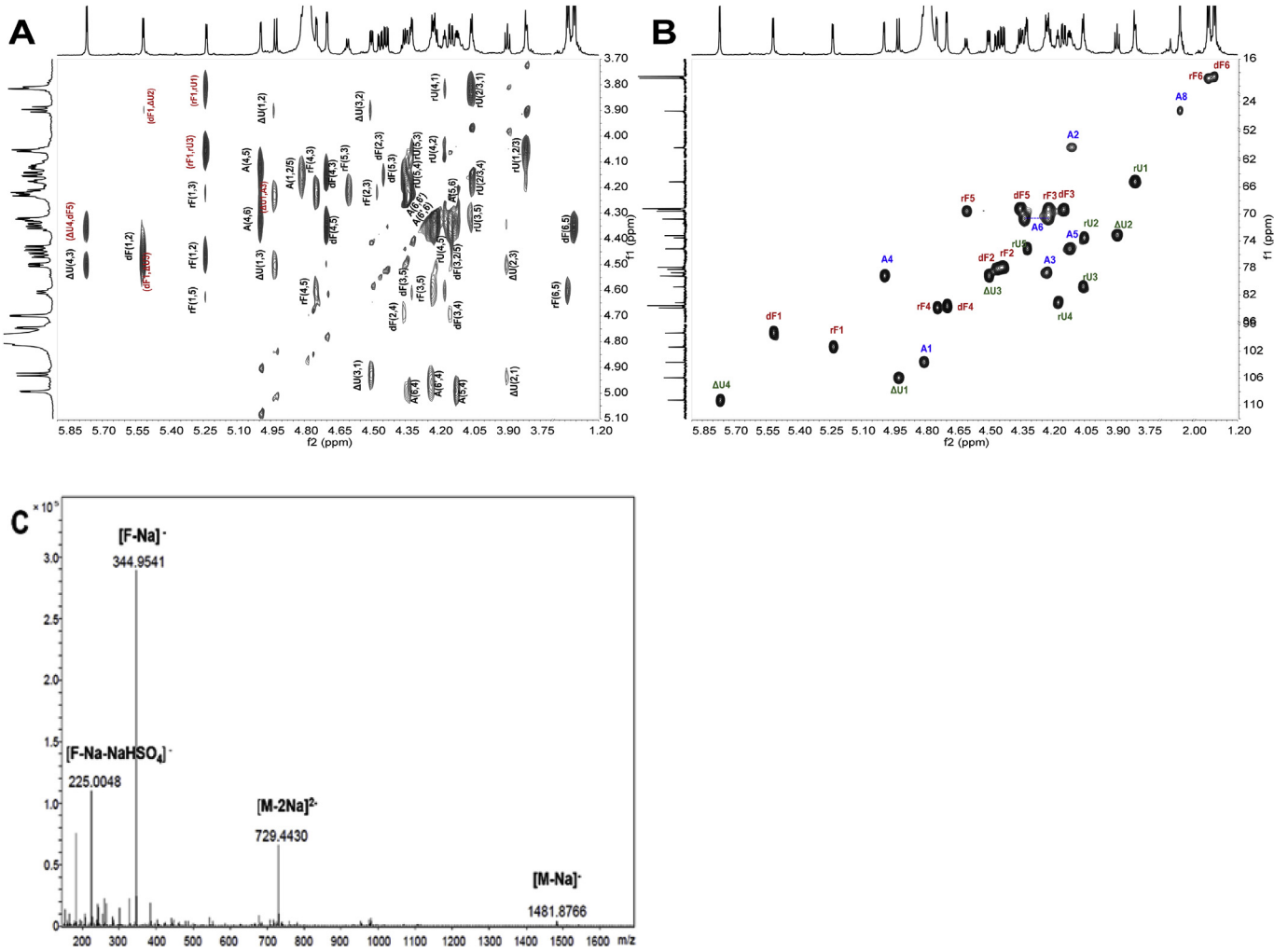


Fig. 3. ^{1}H-^{1}H ROESY (A), ^{1}H-^{13}C HSQC (B) NMR spectra and ESI-Q-TOF MS (C) of compound **7**.

ΔU-β(1 → 3)-*D*-GalNAc$_{4S6S}$-β(1 → 4)-[*L*-Fuc$_{2S4S}$-α(1 → 3)]-*D*-GlcA-β(1 → 3)-*D*-GalNAc$_{4S6S}$-β(1 → 4)-[*L*-Fuc$_{2S4S}$-α(1 → 3)]-*D*-GlcA-ol.

Overall, as shown in Fig. 5, the sequences of these oligosaccharides all contain the repeating trisaccharide unit −{[*L*-Fuc-α(1 → 3)]-*D*-GlcA-β(1 → 3)-*D*-GalNAc-β(1 → 4)-}-. The Fuc side chains and GalNAc residues of this unit are mostly Fuc$_{2S4S}$ and GalNAc$_{4S6S}$, respectively.

Furthermore, another two fractions (compound **6** and peak **I**) were purified from product dFG (Fig. 1), which were much smaller in amount and molecular size compared with compound **7**. The NMR spectra clearly showed that **6** is a trisaccharide (Fig. S7). The complete assignment of its signal peaks is given in Table S2. The signals at 5.528, 3.768 (3.706), and 5.226 ppm were identified as *H*-1 proton of Fuc, GalNAc-ol, ΔU and their *C*-1 chemical shifts were at 100.03, 103.82, and 62.52 ppm, respectively (Fig. S7B). The downfield shifts of *H*-2 (4.448 ppm) and *H*-4 (4.711 ppm) of Fuc, *H*-4 (4.465 ppm) and H-6 (4.192 & 4.35 ppm) of GalNAc-ol indicated that Fuc is sulfated at both the *C*-2 and *C*-4 positions and that GalNAc-ol is sulfated at both the *C*-4 and *C*-6 positions. The linkage types were consistent with those found in compounds **7** and **8** deduced from their ROESY and HMBC spectra (Figs. S7D–E). ESI-Q-TOF-MS analysis revealed an *m*/*z* of 933.9284 for [M − Na]$^{-}$. This result was effectively the same as the calculated value of 933.9322, confirming that the molecular formula of **6** is $C_{20}H_{28}O_{27}NS_4Na_5$ (Fig. S7F, Table S3). Thus, **6** was determined to be the trisaccharide *L*-Fuc$_{2S4S}$-α(1 → 3)-*D*-ΔU-β(1 → 3)-*D*-GalNAc$_{4S6S}$-ol.

Peak I may be a mixture of monosaccharides or their fragment derivatives produced by alkaline degradation during the β-eliminative cleavage according to its ^{1}H NMR spectrum (Fig. S8). From this spectrum, we surmised that some side fucosyl branches should be hydrolysis from the polysaccharides. Signals at ~1.23 ppm were likely to correspond to H-6 signals of free sulfated fucose monosaccharides. Notably, the H-1 of these sulfated fucose monosaccharides were at 3.6 ppm. The data indicated that these monosaccharaides should be further reduced by $NaBH_4$ into fucitol. There were a series of signals at 5.95–5.70 ppm, which could be assigned to olefinic hydrogen. This suggests that the monosaccharides at reducing ends were damaged by a peeling reaction under alkaline conditions, as found in the heparin degradation process [40].

2.4. Occurrence of the peeling reaction during β-eliminative depolymerization

Some studies have revealed that if a polysaccharide contains 1 → 3 or 1 → 4 glycosidic bonds, its reducing ends are prone to incur a peeling reaction under alkaline conditions. This results in peeling off its reducing ends of the sugar group [41]. According to the chemical structure of these oligosaccharides (Fig. 5), we speculated that a peeling reaction for the depolymerized FG products

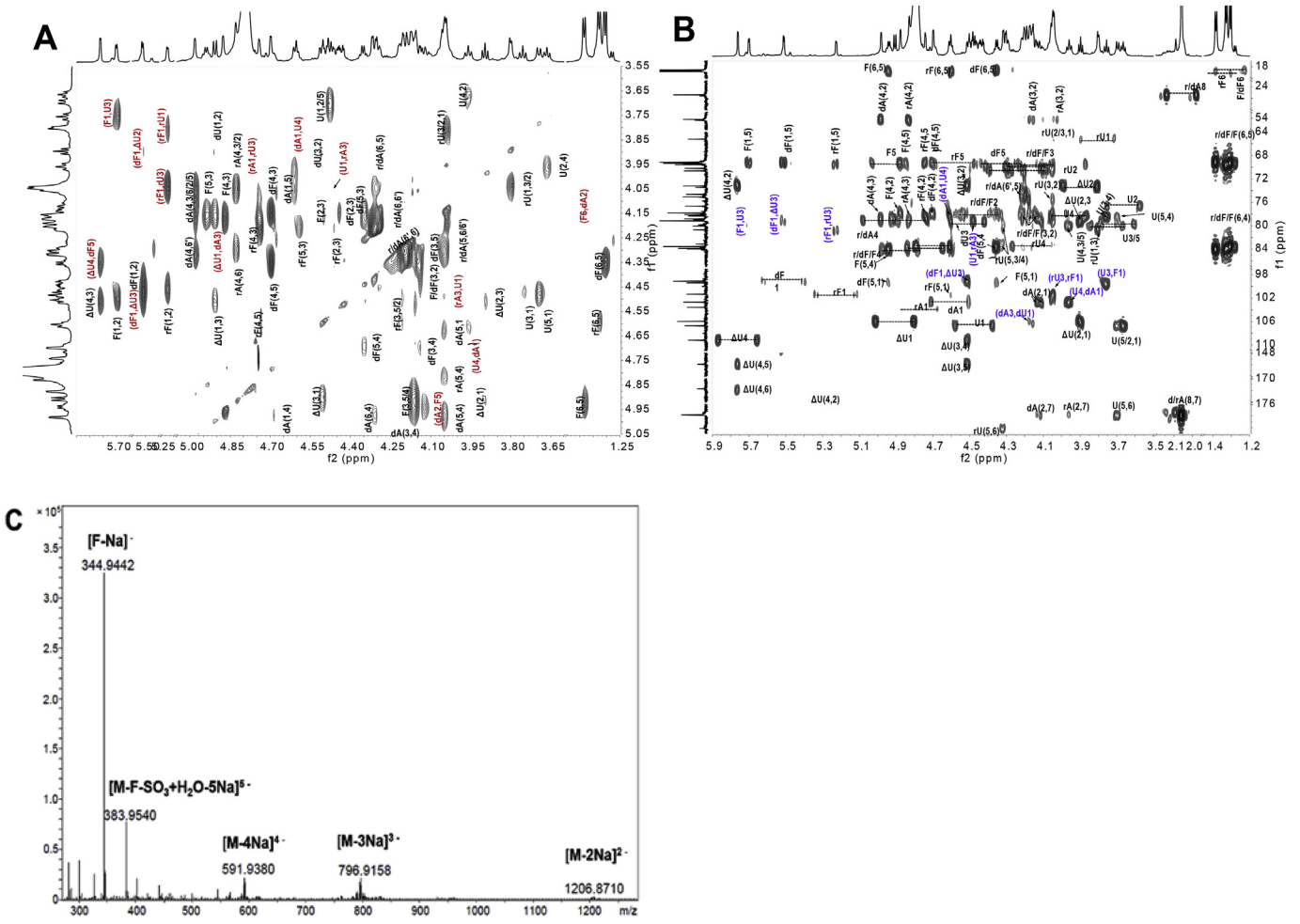


Fig. 4. ^{1}H-^{1}H ROESY (A), ^{1}H-^{13}C HMBC (B) NMR spectra and ESI-Q-TOF MS (C) of compound **8**.

occurred after β-eliminative depolymerization (Scheme 2A). The new reducing ends can be reduced to the corresponding form of sugar alcohol by $NaBH_4$ [42]. Thus, the reducing-end residues of our obtained oligosaccharides **7** and **8** were deduced to be GlcA-ol (Scheme 2B).

The peeling reaction is a base-catalyzed elimination reaction, resulting in the stepwise loss of the monosaccharide residue from the reducing-end of carbohydrate polymers. Peeling reactions contribute to the degradation of oligosaccharides released from glycopeptides and glycoproteins using hydrozinolysis or β-elimination [43–48]. The degradation is the most facile for reducing-end monosaccharides substituted at the 3-position, due to the loss of substituent [41]. The 1, 4-linked glycans require harsh conditions to induce structural rearrangements before undergoing peeling reactions [47,48]. As with our results, the reducing-end monosaccharides of oligosaccharides **7** and **8** are GlcA. We reasoned that such a peeling reaction observed for GalNAc saccharides could have occurred. This elimination would create a double bond at the *C*2–*C*3 position of the GalNAc, which further sets the stage for the peeling (elimination) at the *C*-3 position under basic conditions (Scheme 2A). In the ^{1}H NMR spectrum of peak I (Fig. S8), the *H*-1 of these sulfated fucose monosaccharides were at 3.6 ppm, indicating that these monosaccharaides should be further reduced by $NaBH_4$ into fucitol. Further, a series of signals at 5.95–5.70 ppm could be assigned to olefinic hydrogen [40]. Thus, the β-elimination was assumed to occur in the aldehyde of the reducing end and directly yields the unsaturated species. Future research could characterize and identify these products.

2.5. Deduction of the FG structure

Overall, the Fuc side chain in **6**~**8** only exists as a monosaccharide and only attaches to GlcA through an α(1 → 3) linkage, a di- or trisaccharide side chain, which was not found, either any Fuc side chain linked to GalNAc. Based on the characteristic structures of **6**~**8**, and considering that GlcA-ols are produced by the peeling reaction, we concluded that the cleavage is highly dependent on the functional group and does not affect the substituted Fuc or sulfate. The structure of native FG **1** can be rationally deduced to be composed of repeating units of *D*-GlcA-β(1 → 3)-*D*-GalNAc-β(1 → 4) in the backbone and a monosaccharide Fuc side chain connected to GlcA through an α(1 → 3) glycosidic bond.

Research about FG for over thirty years, with increasingly in-depth study of their structure, leads to some consensus. For example, FG has the backbone of chondroitin sulfate E, i.e. →4)-GlcA-β(1 → 3)-GalNAc-β(1→, and sulfation substitutes in *C*-6 or *C*-4 of GalNAc [1]. However, two main divergences remain to be solved as shown in Fig. 6. Firstly, whether the side chains comprised of monosaccharide (Fig. 6A) [12,16,19–22], disaccharide (Fig. 6B) [14,17] or trisaccharide (Fig. 6C) [10,15]. Secondly, whether the side chains are linked to *C*-3 of GlcA or *C*-4 or/and *C*-6 of GalNAc (Fig. 6A) [17].

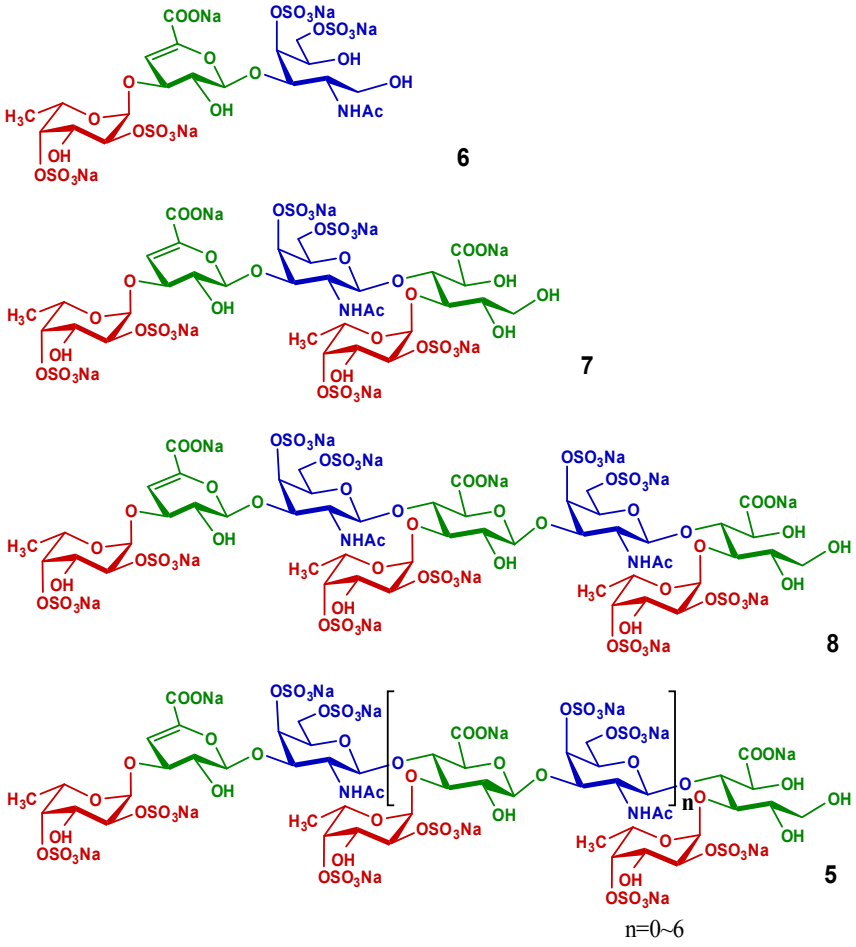


Fig. 5. Chemical structures of FG-derived oligosaccharides (**6**~**8**) and depolymerized FG **5**. The $\Delta^{4,5}$ unsaturated glucuronic acid presents in the non-reducing end of **6** (trisaccharide), **7** (pentasaccharide), **8** (octasaccharide) and dFG **5**.

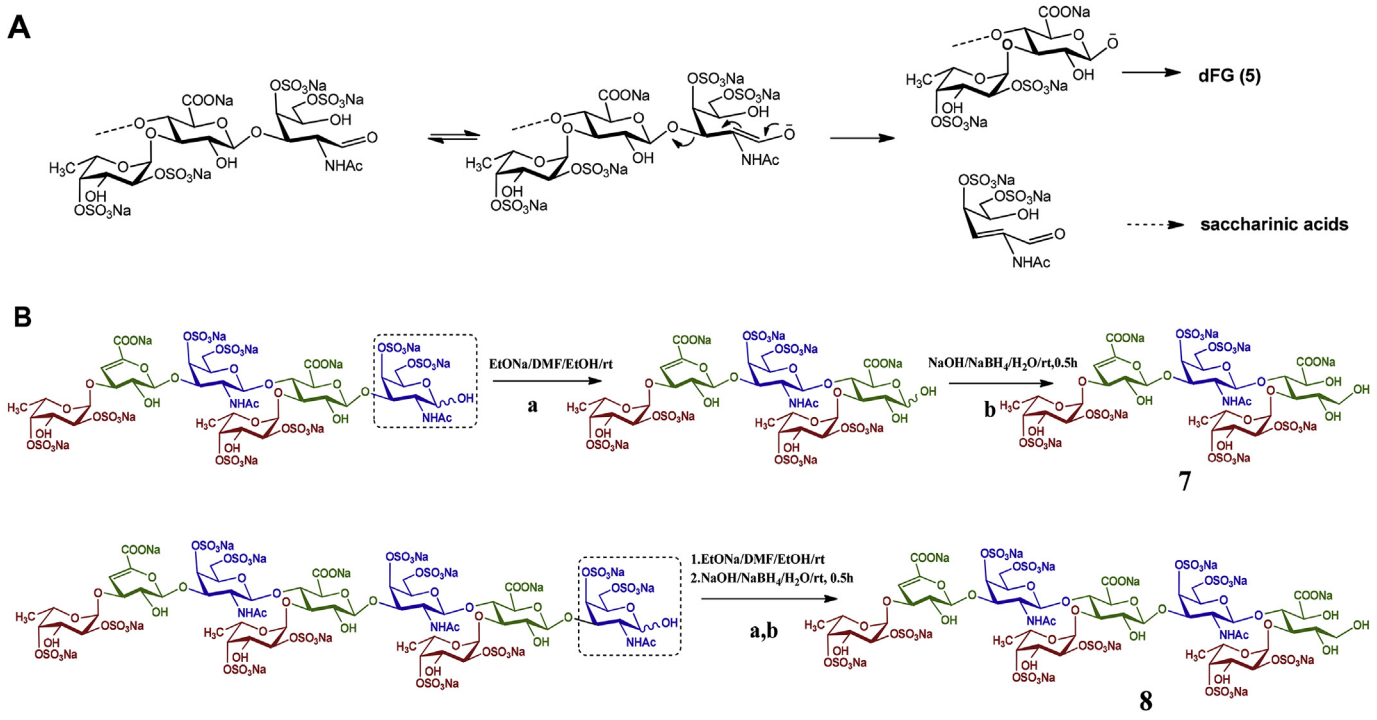
Apart from the structural differences among FGs from various sea cucumber species, inconsistent conclusions about the structure analyzed by various research groups might be due to interference factors. For instance, disparities could arise from the incompletion of de-fucoses by partial acid hydrolysis [11], the removal of fucoses in the process of desulfuration by solvolysis [10,11], and the possibility of removing sulfate ester and the substituent groups during the process of methylation [17]. Therefore, further studies are required to develop new methods to clarify the chemical structure of FGs.

In this paper, taking a FG from the sea cucumber *S. horrens* as an example, a series of oligosaccharides were prepared by a selective glycosidic bond cleavage method and then obtained by a gel purification method. The structures of these oligosaccharides were analyzed by NMR and ESI-Q-TOF MS. Because this depolymerization method is highly selective to the glycosidic bond, the structures of these oligosaccharides showed a certain regularity (Figs. 5, 6D). This provides direct evidence of the chemical structure of FG, and then the chemical structure of the macromolecule FG should be inferred. In our strategy, characteristic unsaturated uronic acid was produced by the selective depolymerization method, without affecting other functional groups such as sulfate ester. Thus, the chemical structure of the complex FG inferred from structures of oligosaccharides should be highly reliable. More importantly, this new reference is conducive to studying of the structure of more complex FGs because this strategy also is applicable in studies of FGs from other material sources. Although the separation process of oligosaccharides is not easy, the obtained oligosaccharides can be used not only to clarify the structure of the parent compound FG, but also to study of their structure-activity relationships.

2.6. Anticoagulant activities of oligosaccharides

The anticoagulant activities of FG **1**, dFG **5** and compounds **6**~**8** were evaluated by plasma clotting assays These include using activated partial thromboplastin time (APTT), prothrombin time (PT) and thrombin time (TT) of plasma clotting assays (Table 1). Such assays are routinely used to determine the ability of products to inhibit blood clotting through the intrinsic, extrinsic and common pathways of the coagulation cascade, respectively [6,49]. The concentrations of **1**, **5** and **7**~**8** needed to double the APTT were between 2.27 and 14.4 μg/mL ($p < 0.01$), indicating that these compounds have potent intrinsic anticoagulant activities, more potent than or similar to LMWH as a positive control. Notably, **6** had negligible effects on APTT. As for PT and TT, no significant influences were observed in **5**~**8** at concentration as high as 128 μg/ml ($p > 0.05$). This result means that they have no or little effect on the extrinsic and common coagulation pathways.

We further found that **1** and **5**~**8** exhibit potent activity for inhibiting intrinsic FXase and have no or weak antithrombin (AT)-dependent inhibition for FXa and FIIa, and that **6** exhibits no inhibition for both FXase and these factors (Table 1, Fig. 7). Compound **1**



Scheme 2. (A) Proposed mechanism for the peeling reaction during β-eliminative depolymerization, (B) proposed process of FG-derived oligosaccharides **7** and **8** with GlcA-ol at the reducing end, (a) peeling reaction under strong alkaline condition and (b) the reduction of glucuronic acid with $NaBH_4$.

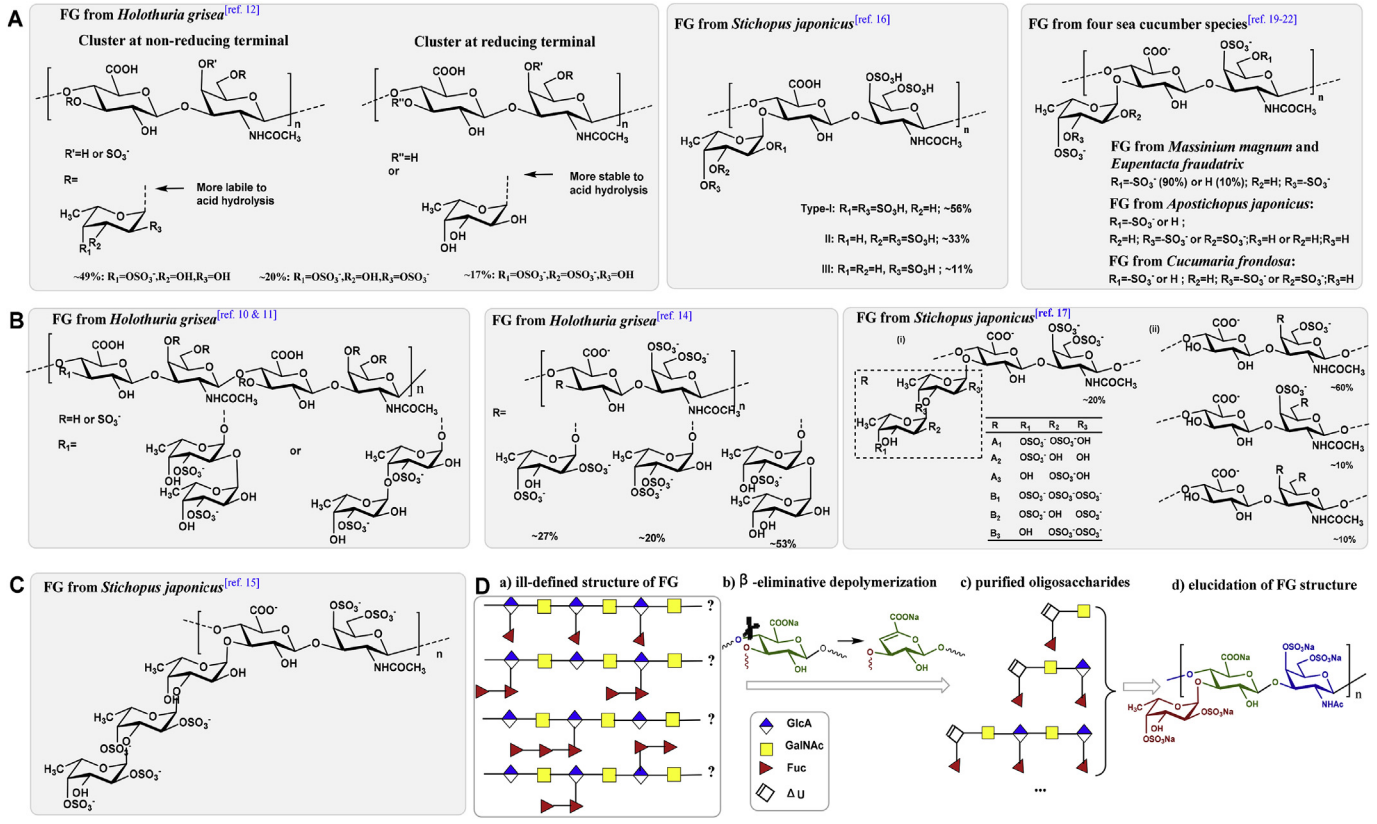


Fig. 6. Proposed structures of FGs in the literature from different sea cucumber species (A, B, C) and elucidation of precise FG structure from *Stichopus horrens* (D).

Table 1
Anticoagulant activities and effects of FG **1**, dFG **5** and compounds **6**~**8** on coagulation factors.

Compd.	Mw[a] (Da)	APTT[b] (μg/mL)	PT[b] (μg/mL)	TT[b] (μg/mL)	Anti-FXase[c] (IC_{50}, ng/mL)	Anti-FIIa/AT[c] (IC_{50}, ng/mL)	Anti-FXa/AT[c] (IC_{50}, ng/mL)
1	46280	2.27**	>128	8.88**	13.6 ± 1.4	344 ± 18	2584 ± 232
5	7210	3.64**	>128	>128	13.5 ± 0.3	1194 ± 66	10629 ± 368
6	956 (dp3)	>128	>128	>128	2319.8 ± 167.9	NA[d]	NA[d]
7	1504 (dp5)	106.90*	>128	>128	1737 ± 539	NA[d]	NA[d]
8	2459 (dp8)	14.43**	>128	>128	243 ± 32	NA[d]	NA[d]
LMWH[e]	4500–5500	6.18**	>128	4.61**	64 ± 14	49.1[f]	26.6[f]

[a] The Mws of FG **1** and **5** were determined by HPGPC. The Mws of compounds **6**~**8** were calculated based on their MS spectra.
[b] The activity of agents to prolong APTT, PT or TT is expressed by the concentration of each agent (μg/mL) that is required to double the APTT, PT or TT. $^{*}p < 0.05$, $^{**}p < 0.01$ *vs.* a negative control (Tris-HCl buffer).
[c] IC_{50} value, the concentration of each agent required to inhibit 50% of protease activity.
[d] NA, inhibition ratio was less than 10% at concentration as high as 2000 ng/mL.
[e] LMWH, a low-molecular-weight heparin (Enoxaparin, 0.4 mL × 4000 AXaIU), was used as a positive control from Sanofi-Aventis (France).
[f] Data from Ref. [7].

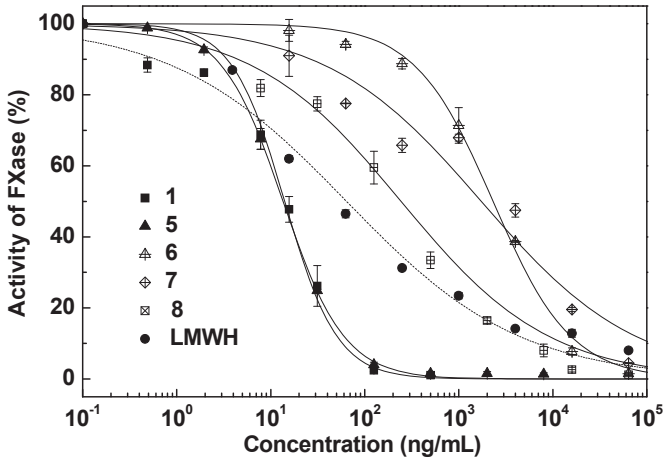


Fig. 7. Anti-FXase activities of compounds **1**, **5**~**8**.

displayed strong AT-dependent anti-FIIa activity, consistent with our previous reports [7]. In contrast, this effect was not observed in **6**~**8** at concentrations as high as 2000 ng/mL. Remarkably, **8** displayed much stronger anti-FXase activity, which was higher than its anti-FXa and anti-FIIa activity in the presence of AT. This indicated that its anticoagulant mechanisms are significantly different from those of heparin-like drugs.

Since intrinsic factor Xase is the last target enzyme for inhibition of the intrinsic coagulation pathway [50,51], it is becoming recognized as a prime target for developing safer anticoagulants with potential physiological and therapeutic applications [3]. Our findings show that structurally novel FG-derived oligosaccharides possess potent factor Xase inhibitory activity, conferring special interest as potential candidate drugs. However, the yields of purified products are low, and a large-scale preparation is difficult to be achieved. Thus, synthesis of these FG-derived oligosaccharides merits further work by synthetic chemists in order to produce these novel oligosaccharides in commercial volumes.

3. Conclusions

Uncertainty of the detailed structures of native FG has endured for over 30 years [1,10–14]. In this study (Fig. 6D), using a FG from the sea cucumber *S. horrens*, we used the selective β-elimination of glycosidic bonds to purify some highly regular oligosaccharides from the depolymerization products. Structural analysis of the oligosaccharides further confirmed the precise structure of the FG: its backbone is →4)-GlcA-β(1 → 3)-GalNAc$_{4S6S}$-β(1→, and a monosaccharide Fuc is linked to GlcA through α(1 → 3) glycosidic bonds. This depolymerization method is highly selective to the glycosidic bond and the structures of the obtained oligosaccharides show a certain regularity. Thus, this method should be applicable for analyzing the structures of other glycosaminoglycans, including FGs from other sea cucumber species.

FG-derived oligosaccharides exhibit strong anticoagulant activity by intrinsic FXase inhibition, and the anticoagulant mechanisms are significantly different from those of heparin-like drugs. The intrinsic tenase complex, which comprises factor VIIIa and factor IXa and converts factor X to FXa, is the last and rate-determining enzyme step along the intrinsic coagulation cascade. Developing inhibitors of this enzyme complex is thus important to meet clinical demand [4]. Therefore, the FG-derived homogeneous oligosaccharides offer promise for elucidating the structure-activity relationship of FG and for developing a new class of anticoagulant agents as intrinsic FXase inhibitors.

4. Experimental section

4.1. Materials

Dried sea cucumbers *Stichopus horrens* were purchased in seafood markets in Sanya of the Hainan province of China. Amberlite FPA98Cl ion exchange resin was purchased from Rohm and Haas Company (USA). Bio-Gel P-2/P-6 and P-10 columns were from Bio-Rad Laboratories and Sephadex G-10/G-25 were from GE Healthcare Life Sicences (USA). Benzyl chloride was from Aladdin chemistry Co. Ltd. Benzethonium chloride was from Yangzhou Aoxin Chemical factory (China). Deuterium oxide (D_2O, 99.9% Atom D) was obtained from Sigma-Aldrich. LMWH (Enoxaparin, 0.4 ml × 4000 AXaIU) was from Sanofi-Aventis (France). The activated partial thromboplastin time (APTT), prothrombin time (PT), thrombin time (TT) reagents, and standard human plasma were from Teco Medical (Germany). Biophen FVIII: C kit, Biophen Heparin Anti-IIa kits, and Biophen Anti-Xa kits were all from Hyphen Biomed (France). Human factor VIII was from Bayer HealthCare LLC. All other chemicals were of reagent grade and obtained commercially.

4.2. Extraction and purification of native FG

Crude polysaccharide from the body wall of the sea cucumber *S. horrens* were prepared using methods previously described [7,25]. Briefly, the tissue of the dried body wall (300 g) was digested by 1% papain for 6 h and released by 0.25 M sodium hydroxide for 2 h. The crude polysaccharide was obtained and further purified by strong anion-exchange chromatography using FPA98 resin

(4 cm × 45 cm), and the sample was eluted successively with 2 BV distilled water, 0.5 M, 1.0 M and 2 M NaCl solution. Each fraction eluted by water and different gradient NaCl solutions was collected, dialyzed and lyophilized. The purity of each fraction was determined by high-performance gel permeation chromatography (HPGPC) using a Agilent technologies 1260 series (Agilent Co., USA) apparatus with differential refraction detector, equipped with a Shodex OH-pak SB-804 HQ column (8 mm × 300 mm). Chromatographic conditions and procedures were performed according to the previous method [7].

4.3. Preparation of depolymerized products by β-elimination

A solution of 500 mg benzethonium chloride (HyCl) in 8 mL H_2O was added to the solution of 201.9 mg FG (**1**) in 2.96 mL H_2O under stirring at room temperature, the resulting white precipitate was collected by centrifugation (4000 rpm × 15 min) and then dried by reduced pressure evaporation (40 °C), the yield of FG benzethonium salts (product, **2**) was 0.587 g. Then, the FG benzethonium salts were dissolved in 2.935 mL dimethyl formamide, and 70 μL benzyl chloride was added. After heating this solution at 35 °C for 24 h, FG benzyl ester (product **3**) was obtained. Then 1.002 mL of freshly prepared 0.08 M EtONa/EtOH was added at room temperature. The reaction was kept for 30 min under magnetic stirring, then 4.47 mL of saturated sodium chloride solution and 35.76 mL EtOH were added and the precipitate was collected by centrifugation (4000 rpm × 15 min) (product **4**). In order to remove the benzyl groups, the resulting precipitate was dissolved in 16 mL H_2O, and 0.271 mL freshly prepared 6 M NaOH was added (depolymerized FG product **4**). After 30 min at 25 °C, 61.64 mg sodium borohydride was added, and the reaction was kept for another 30 min under magnetic stirring. After neutralizing with HCl and precipitating by addition of 65.2 mL absolute ethanol, the resulting precipitate was desalted by Sephadex G-25. It was then lyophilized to obtain about 110 mg of the final product (dFG **5**) with a yield of ~55% based on the starting material **1**.

The ratio of sulfate/carboxyl groups was determined by a conductometric method [52]. The molar ratio of sulfate to carboxylate groups of FG is 4:1. The specific optical rotation of 10 mL of 1% (w/v) samples in water was determined at 25 °C, and **5** exhibits an optical rotation $[\alpha]_D$ of −60.7°, which is similar to that of FG (−64.5°). ^{1}H NMR (600 MHz, D_2O) δ 5.760 (1H, d, $H_{\Delta U}$-4, $J_{(3,4)} = 2.48$ Hz), 5.693 (1H, m, H_F-1), 5.508 (1H, d, H_{dF}-1, $J_{(1,2)} = 3.68$ Hz), 5.226 (1H, d, H_{rF}-1, $J_{(1,2)} = 3.44$ Hz), 4.980 (1H, m, H_{dA}-4), 4.909 (1H, m, H_F-5, $J_{(5,6)} = 6.56$ Hz), 4.899 (1H, d, $H_{\Delta U}$-1, $J_{(1,2)} = 8.16$ Hz), 4.877 (1H, m, H_F-4), 4.801 (2H, m, H_A-4, H_{rA}-4), 4.757 (1H, m, H_{rA}-1), 4.739 (1H, m, H_{rF}-4), 4.697 (1H, m, H_{dF}-4), 4.595 (1H, m, H_{rF}-5, $J_{(5,6)} = 6.40$ Hz), 4.589 (1H, m, H_{dA}-1), 4.567 (1H, m, H_A-1), 4.502 (1H, dd, $H_{\Delta U}$-3, $J_{(3,4)} = 2.48$ Hz, $J_{(2,3)} = 8.16$ Hz), 4.490 (1H, m, H_F-2), 4.463 (1H, d, H_U-1, $J_{(1,2)} = 7.52$ Hz), 4.463 (1H, m, H_{rF}-2), 4.432 (1H, m, H_{dF}-2), 4.350 (1H, m, H_{dF}-5, $J_{(5,6)} = 6.56$ Hz), 4.313 (1H, m, H_{rU}-5), 4.272 (1H, m, H_{dA}-6), 4.269 (2H, m, H_A-6, H_{rA}-6), 4.207 (1H, m, H_{rF}-3, $J_{(3,4)} = 2.96$ Hz), 4.201 (1H, m, H_{dA}-6′), 4.175 (2H, m, H_A-6′, H_{rA}-6′), 4.162 (1H, m, H_{rU}-4), 4.159 (1H, m, H_F-3), 4.151 (1H, m, H_{dA}-3), 4.148 (1H, m, H_{dF}-3, $J_{(3,4)} = 2.88$ Hz), 4.118 (1H, m, H_{dA}-2, $J_{(2,3)} = 9.04$ Hz), 4.050 (2H, m, H_{rU}-2, H_{rU}-3), 4.049 (2H, m, H_{rA}-2, H_{rA}-3), 4.046 (3H, m, H_{dA}-5, H_A-2, H_{rA}-5), 3.968 (1H, m, H_A-5), 3.948 (1H, dd, H_U-4, $J_{(3,4)} = 8.48$ Hz, $J_{(4,5)} = 7.60$ Hz), 3.939 (1H, m, H_A-3), 3.895 (1H, dd, $H_{\Delta U}$-2, $J_{(1,2)} = J_{(2,3)} = 8.16$ Hz), 3.799 (2H, d, H_{rU}-1, $J_{(1,2)} = 6.16$ Hz), 3.740 (1H, dd, H_U-3, $J_{(2,3)} = 8.16$ Hz, $J_{(3,4)} = 8.48$ Hz), 3.682 (1H, d, H_U-5, $J_{(4,5)} = 7.60$ Hz), 3.648 (1H, dd, H_U-2, $J_{(1,2)} = 7.52$ Hz, $J_{(2,3)} = 8.40$ Hz), 2.061 (3H, s, H_{dA}-8), 2.053 (6H, s, H_A-8, H_{rA}-8), 1.363 (3H, s, H_F-6), 1.327 (3H, s, H_{rF}-6), 1.304 (3H, s, H_{dF}-6); ^{13}C NMR (150 MHz, D_2O) δ180.15 (C_{rU}-6), 177.86 (C_{dA}-7, C_A-7, C_{rA}-7), 177.83 (C_U-6), 171.73 ($C_{\Delta U}$-6), 149.93 ($C_{\Delta U}$-5), 109.19 ($C_{\Delta U}$-4), 106.94 (C_U-1), 105.99 ($C_{\Delta U}$-1), 103.56 (C_{rA}-1), 102.66 (C_{dA}-1, C_A-1), 101.36 (C_{rF}-1), 99.33 (C_F-1), 99.11 (C_{dF}-1), 83.96 (C_F-4), 83.90 (C_{rF}-4), 83.57 (C_{dF}-4), 83.09 (C_{rU}-4), 80.72 (C_{rU}-3), 79.70 (C_U-5), 79.69 (C_A-3), 79.25 (C_{rA}-3), 79.16 (C_{dA}-4, C_A-4, C_{rA}-4), 79.13 (C_U-3), 79.07 (C_{dA}-3), 78.87 ($C_{\Delta U}$-3), 78.29 (C_U-4), 78.16 (C_{rF}-2), 77.91 (C_F-2), 77.84 (C_{dF}-2), 76.77 (C_U-2), 75.08 (C_{rU}-5, C_{rA}-5), 74.67 (C_{dA}-5), 74.60 (C_A-5), 73.47 (C_{rU}-2), 73.22 ($C_{\Delta U}$-2), 70.66 (C_{rA}-6), 70.21 (C_{dA}-6), 70.12 (C_A-6), 69.58 (C_{rF}-5), 69.40 (C_F-3), 69.33 (C_{dF}-3), 69.23 (C_{rF}-3), 69.15 (C_{dF}-5, C_F-5), 65.21 (C_{rU}-1), 54.37 (C_{dA}-2),54.21 (C_{rA}-2), 54.12 (C_A-2), 25.51 (C_A-8, C_{rA}-8), 25.43 (C_{dA}-8), 18.83 (C_{rF}-6), 18.71 (C_F-6), 18.55 (C_{dF}-6).

4.4. Preparation of oligosaccharides

Approximately 1 g of **5** was prepared from native FG by β-elimination as described above. The solution of **5** was filtered through a 0.45 μm syringe filter and fractionated on a Bio-Gel P-10 column (medium, 2 cm × 200 cm) eluted with 0.2 M NaCl solution at a flow rate of 16.2 mL/h or on a Bio-Gel P-6 column (fine, 2 cm × 140 cm) eluted with 0.2 M NaCl solution at a flow rate of 9 mL/h. Approximately 3.2 mL fractions were collected and fractions obtained in time sequence were plotted versus 234 nm ultraviolet absorbance. Fractions of each peak containing oligosaccharides with the same molecular size as the single peaks showed were pooled. The resulting oligosaccharides were further purified by rechromatography on the Bio-Gel P-10 and P-6 columns several times to ensure the homogeneity of molecular size as determined by analytical HPGPC. The final pure oligosaccharides were desalted by Sephadex G-10/Bio-Gel P-2 column and lyophilized [6].

4.5. Physicochemical characterization

The weight-average molecular mass (M_w), number-average molecular mass (M_n) and molecular weight distribution (M_w/M_n) of native **1** and **5** (dFG) were determined by high-performance gel permeation chromatography (HPGPC) using an Agilent technologies 1260 series (Agilent Co., USA) apparatus equipped with a Shodex OH-pak SB-804 HQ column (8 mm × 300 mm). Chromatographic procedures and conditions were performed according to a previously described method [7]. Specifically, for molecular weight determination, a standard curve was calibrated by standard D-series Dextrans (D-0, 2, 3, 4, 5, 6, 7 and 8). Molecular weight calculations were performed by a GPC software, version B01.01 (Agilent Co., USA) and corrected by the standard product (low molecular weight FG with Mw 11.58 kDa).

4.6. HPGPC analysis of oligosaccharides

50 μL of 2 mg/mL oligosaccharide samples were size fractionated by HPLC with a Superdex Peptide 10/300 GL column with 0.2 M NaCl solution as the eluent at a flow rate of 0.4 mL/min monitored by a differential refraction detector. Compounds **6**–**8** were analyzed as described previously [6].

4.7. ESI-Q-TOF-MS analysis

Negative-ion ESI-MS was conducted on a 6540 UHD Accurate-Mass Q-TOF LC/MS spectrometer (Agilent Technologies, USA). Data analysis was performed using Agilent Mass-Hunter Workstation Data Acquisition (Agilent Technologies, USA). The MS spectrometric conditions were as follows: ESI in negative ion mode, capillary voltage of 3500 V, nebulizer pressure of 30 psig, drying gas flow rate of 8 L/min, and drying gas temperature of 350 °C. The mass spectra of the oligosaccharides were acquired in scan mode (m/z scan range 200–2000).

4.8. UV, IR and NMR analysis

Ultraviolet (UV) analysis was performed with a UV-2450 spectrophotometer (Shimadzu, Japan) in the wavelength range of 190–800 nm. The UV scan spectrum of native FG showed no obvious absorbance at 260 nm or 280 nm except the strong absorption at about 210 nm. This result demonstrated that this purified FG was not contaminated with proteins or nucleic acids. The low molecular weight product dFG (**5**) showed special maximum absorption (λ_{max}) at 234 nm, which indicates the presence of the $\Delta^{4,5}$ unsaturated glucuronic acid derived from β-eliminative cleavage of the glycosidic linkage of the FG.

The FT-IR spectrum (KBr pellets) of the FG was recorded by a Tensor-27 (Bruker, Germany) in a range of 400–4000 cm^{-1} at room temperature. The FT-IR spectrum showed several bands corresponding to sulfate ester: the signal peaks at 1244 and 851 cm^{-1} are produced by the stretching vibration of S=O of sulfate and the bending vibration of C-O-S of sulfate in an axial position, respectively. The signals at 3443 and 1031 cm^{-1} are from the stretching vibration of O-H and C-O, respectively. In addition, signals at 1632 cm^{-1} were ascribed to the asymmetric stretching vibration of C=O of N-acetyl galactosamine and glucuronic acid; 2941 cm^{-1}, the stretching vibration of C-H; 1412 cm^{-1}, the symmetric stretching vibration of COO^- of glucuronic acid and the stretching vibration of C-O within COOH.

The structure analysis of FG, **5** and compounds **6**~**8** by NMR analyses were performed in D_2O at 298 K with a Bruker Advance 600 or 800 MHz spectrometer equipped with a $^{13}C/^{1}H$ dual probe in FT model [7]. All spectra were recorded with HOD suppression by presaturation. ^{1}H-^{1}H COSY, TOCSY, NOESY, ^{1}H-^{13}C HSQC, and HMBC spectra were recorded using state–time proportional phase incrementation for quadrature detection in the indirect dimension. All chemical shifts were relative to internal trimethylsilyl-propionic acid (TSP). All samples were previously dissolved in deuterium oxide (D_2O, 99.9% D) and lyophilized three times to replace exchangeable protons with D_2O. The lyophilized samples were then dissolved in D_2O at a concentration of 20 g/L. The desalted, pure oligosaccharides obtained were water-soluble white powders (HPLC purity > 98%) (compounds **6**~**8**).

Trisaccharide *L*-Fuc$_{2S4S}$-α(1 → 3)-*D*-ΔU-β(1 → 3)-*D*-GalNAc$_{4S6S}$-ol (6): ^{1}H NMR (800 MHz, D_2O) δ 5.888 (1H, d, $H_{\Delta U}$-4, $J_{(3,4)} = 4.00$ Hz), 5.528 (1H, d, H_F-1, $J_{(1,2)} = 4.64$ Hz), 5.226 (1H, d, $H_{\Delta U}$-1, $J_{(1,2)} = 5.36$ Hz), 4.711 (1H, dd, H_F-4, $J_{(3,4)} = 5.36$ Hz), 4.465 (1H, m, H_{rA}-4), 4.452 (1H, m, H_{rA}-3, $J_{(2,3)} = 7.52$ Hz), 4.448 (1H, dd, H_F-2, $J_{(1,2)} = 4.64$ Hz, $J_{(2,3)} = 10.88$ Hz), 4.401 (1H, m, H_{rA}-5, $J_{(5,6/6')} = 8.08$ Hz, 4.56 Hz), 4.350 (1H, m, H_{rA}-6, $J_{(6,6')} = 10.24$ Hz), 4.338 (1H, m, H_F-5, $J_{(5,6)} = 6.64$ Hz), 4.310 (1H, dd, $H_{\Delta U}$-3, $J_{(2,3)} = 4.40$ Hz, $J_{(3,4)} = 4.00$ Hz), 4.288 (1H, dd, H_{rA}-2, $J_{(2,3)} = 7.52$ Hz), 4.192 (1H, m, H_{rA}-6, $J_{(6,6')} = 10.24$ Hz), 4.150 (1H, dd, $H_{\Delta U}$-2, $J_{(1,2)} = 5.36$ Hz, $J_{(2,3)} = 4.40$ Hz), 4.048 (1H, dd, H_F-3, $J_{(2,3)} = 10.88$ Hz, $J_{(3,4)} = 5.36$ Hz), 3.768 (1H, m, H_{rA}-1, $J_{(1,1')} = 11.04$ Hz), 3.706 (1H, m, H_{rA}-1′, $J_{(1/1',2)} = 6.48$ Hz, 8.40 Hz), 2.052 (3H, s, H_{rA}-8), 1.315 (3H, d, H_F-6, $J_{(5,6)} = 6.64$ Hz); ^{13}C NMR (200 MHz, D_2O) δ177.20 (C_{rA}-7), 171.76 ($C_{\Delta U}$-6), 148.40 ($C_{\Delta U}$-5), 108.20 ($C_{\Delta U}$-4, $J_{(H,C)} = 166.4$ Hz), 103.82 ($C_{\Delta U}$-1, $J_{(H,C)} = 168.0$ Hz), 100.03 (C_F-1, $J_{(H,C)} = 168.8$ Hz), 83.60 (C_F-4, $J_{(H,C)} = 151.2$ Hz), 79.91 (C_{rA}-4, $J_{(H,C)} = 157.6$ Hz), 77.73 (C_F-2, $J_{(H,C)} = 147.2$ Hz), 77.58 (C_{rA}-3, $J_{(H,C)} = 147.2$ Hz), 76.95 ($C_{\Delta U}$-3), 72.70 (C_{rA}-6, $J_{(H,C)} = 151.2$ Hz), 71.36 ($C_{\Delta U}$-2), 69.29 (C_F-3, $J_{(H,C)} = 158.4$ Hz), 69.88 (C_{rA}-5), 69.19 (C_F-5), 62.52 (C_{rA}-1, $J_{(H,C)} = 151.2$ Hz, 145.6 Hz), 54.16 (C_{rA}-2), 25.01 (C_{rA}-8, $J_{(H,C)} = 130.4$ Hz), 18.59 (C_F-6, $J_{(H,C)} = 121.6$ Hz); HRMS (ESI-Q-TOF) *m/z*: $[M - Na]^-$ calcd for $C_{20}H_{28}O_{27}NS_4Na_4$ 933.9322; Found 933.9284.

Pentasaccharide [*L*-Fuc$_{2S4S}$]-α(1 → 3)-*D*-ΔU-β(1 → 3)-*D*-GalNAc$_{4S6S}$-β(1 → 4)-[*L*-Fuc$_{2S4S}$]-α(1 → 3)-*D*-GlcA-ol (7): ^{1}H NMR (800 MHz, D_2O) δ 5.768 (1H, d, $H_{\Delta U}$-4, $J_{(3,4)} = 2.88$ Hz), 5.516 (1H, d, H_{dF}-1, $J_{(1,2)} = 3.84$ Hz), 5.236 (1H, d, H_{rF}-1, $J_{(1,2)} = 3.76$ Hz), 4.993 (1H, m, H_A-4, $J_{(3,4)} = 2.88$ Hz), 4.928 (1H, d, $H_{\Delta U}$-1, $J_{(1,2)} = 8.48$ Hz), 4.805 (1H, d, H_A-1, $J_{(1,2)} = 8.16$ Hz), 4.746 (1H, m, H_{rF}-4, $J_{(3,4)} = 2.80$ Hz), 4.699 (1H, m, H_{dF}-4, $J_{(3,4)} = 2.96$ Hz), 4.626 (1H, dd, H_{rF}-2, $J_{(1,2)} = 3.76$ Hz, $J_{(2,3)} = 10.56$ Hz), 4.606 (1H, m, H_{rF}-5, $J_{(5,6)} = 6.56$ Hz), 4.502 (1H, dd, $H_{\Delta U}$-3, $J_{(2,3)} = 7.60$ Hz, $J_{(3,4)} = 2.88$ Hz), 4.436 (1H, dd, H_{dF}-2, $J_{(1,2)} = 3.84$ Hz, $J_{(2,3)} = 10.48$ Hz), 4.355 (1H, m, H_{dF}-5, $J_{(5,6)} = 6.56$ Hz), 4.336 (1H, m, H_A-6, $J_{(6,6')} = 10.64$ Hz), 4.321 (1H, d, H_{rU}-5, $J_{(4,5)} = 3.60$ Hz), 4.235 (1H, dd, H_A-3, $J_{(2,3)} = 8.48$ Hz, $J_{(3,4)} = 2.88$ Hz), 4.222 (1H, dd, H_{rF}-3, $J_{(2,3)} = 10.56$ Hz, $J_{(3,4)} = 2.80$ Hz), 4.220 (1H, m, H_A-6′, $J_{(6,6')} = 10.64$ Hz), 4.175 (1H, dd, H_{rU}-4, $J_{(3,4)} = 4.56$ Hz, $J_{(4,5)} = 3.60$ Hz), 4.148 (1H, dd, H_{dF}-3, $J_{(2,3)} = 10.48$ Hz, $J_{(3,4)} = 2.96$ Hz), 4.120 (1H, m, H_A-5, $J_{(5,6/6')} = 7.84$ Hz, 3.36 Hz), 4.109 (1H, dd, H_A-2, $J_{(1,2)} = 8.16$ Hz, $J_{(2,3)} = 8.48$ Hz), 4.060 (1H, dd, H_{rU}-3, $J_{(2,3)} = 5.04$ Hz, $J_{(3,4)} = 4.56$ Hz), 4.054 (1H, dd, H_{rU}-2, $J_{(1,2)} = 4.04$ Hz, $J_{(2,3)} = 5.04$ Hz), 3.896 (1H, dd, $H_{\Delta U}$-2, $J_{(1,2)} = 8.48$ Hz, $J_{(2,3)} = 7.60$ Hz), 3.813 (1H, d, H_{rU}-1, $J_{(1,2)} = 4.04$ Hz), 2.060 (3H, s, H_A-8), 1.337 (3H, d, H_{rF}-6, $J_{(5,6)} = 6.56$ Hz), 1.308 (3H, d, H_{dF}-6, $J_{(5,6)} = 6.56$ Hz); ^{13}C NMR (200 MHz, D_2O) δ 180.18 (C_{rU}-6), 177.74 (C_A-7), 171.74 ($C_{\Delta U}$-6), 149.84 ($C_{\Delta U}$-5), 109.24 ($C_{\Delta U}$-4), 105.94 ($C_{\Delta U}$-1), 103.63 (C_A-1), 101.38 (C_{rF}-1), 99.30 (C_{dF}-1), 83.91 (C_{rF}-4), 83.58 (C_{dF}-4), 83.13 (C_{rU}-4), 80.79 (C_{rU}-3), 79.18 ($C_{\Delta U}$-3, C_A-4), 78.70 (C_A-3), 78.18 (C_{rF}-2), 77.86 (C_{dF}-2), 75.16 (C_A-5), 75.14 (C_{rU}-5), 73.53 (C_{rU}-2), 73.18 ($C_{\Delta U}$-2), 70.73 (C_A-6), 69.59 (C_{rF}-5), 69.34 (C_{dF}-3), 69.26 (C_{rF}-3), 69.19 (C_{dF}-5), 65.22 (C_{rU}-1), 54.50 (C_A-2), 25.39 (C_A-8), 18.85 (C_{rF}-6), 18.59 (C_{dF}-6); HRMS (ESI-Q-TOF) *m/z*: $[M - Na]^-$ calcd for $C_{32}H_{43}O_{54}NS_6Na_7$ 1481.8822; Found 1481.8766.

Octasaccharide [*L*-Fuc$_{2S4S}$-α(1 → 3)]-*D*-ΔU-β(1 → 3)-*D*-GalNAc$_{4S6S}$-β(1 → 4)-[*L*-Fuc$_{2S4S}$-α(1 → 3)]-*D*-GlcA-β(1 → 3)-*D*-GalNAc$_{4S6S}$-β(1 → 4)-[*L*-Fuc$_{2S4S}$-α(1 → 3)]-*D*-GlcA-ol (8): ^{1}H NMR (800 MHz, D_2O) δ 5.760 (1H, m, $H_{\Delta U}$-4), 5.700 (1H, d, H_F-1, $J_{(1,2)} = 3.68$ Hz), 5.511 (1H, d, H_{dF}-1, $J_{(1,2)} = 3.68$ Hz), 5.226 (1H, d, H_{rF}-1, $J_{(1,2)} = 3.44$ Hz), 4.982 (1H, m, H_{dA}-4), 4.939 (1H, m, H_F-5, $J_{(5,6)} = 6.48$ Hz), 4.906 (1H, d, $H_{\Delta U}$-1, $J_{(1,2)} = 8.48$ Hz), 4.876 (1H, m, H_F-4, $J_{(3,4)} = 2.56$ Hz), 4.829 (1H, m, H_{rA}-4), 4.743 (1H, m, H_{rA}-1), 4.741 (1H, m, H_{rF}-4, $J_{(3,4)} = 2.00$ Hz), 4.698 (1H, m, H_{dF}-4, $J_{(3,4)} = 2.56$ Hz), 4.608 (1H, d, H_{dA}-1, $J_{(1,2)} = 8.32$ Hz), 4.598 (1H, m, H_{rF}-5, $J_{(5,6)} = 6.64$ Hz), 4.510 (1H, dd, $H_{\Delta U}$-3, $J_{(2,3)} = 8.08$ Hz, $J_{(3,4)} = 2.40$ Hz), 4.493 (1H, dd, H_F-2, $J_{(1,2)} = 3.68$ Hz, $J_{(2,3)} = 11.12$ Hz), 4.477 (1H, d, H_U-1, $J_{(1,2)} = 8.40$ Hz), 4.455 (1H, dd, H_{rF}-2, $J_{(1,2)} = 3.44$ Hz, $J_{(2,3)} = 10.48$ Hz), 4.433 (1H, dd, H_{dF}-2, $J_{(1,2)} = 3.68$ Hz, $J_{(2,3)} = 10.56$ Hz), 4.353 (1H, m, H_{dF}-5, $J_{(5,6)} = 6.56$ Hz), 4.311 (1H, m, H_{rU}-5), 4.310 (1H, m, H_{dA}-6, $J_{(6,6')} = 11.52$ Hz), 4.296 (1H, m, H_{rA}-6, $J_{(6,6')} = 11.60$ Hz), 4.211 (1H, dd, H_{rF}-3, $J_{(2,3)} = 10.48$ Hz, $J_{(3,4)} = 2.00$ Hz), 4.190 (1H, m, H_{rA}-6′, $J_{(6,6')} = 11.60$ Hz), 4.173 (1H, m, H_{rU}-4), 4.165 (1H, dd, H_F-3, $J_{(2,3)} = 11.12$ Hz, $J_{(3,4)} = 2.56$ Hz), 4.155 (1H, m, H_{dA}-3, $J_{(2,3)} = 9.64$ Hz), 4.146 (1H, dd, H_{dF}-3, $J_{(2,3)} = 10.56$ Hz, $J_{(3,4)} = 2.56$ Hz), 4.121 (1H, dd, H_{dA}-2, $J_{(1,2)} = 8.32$ Hz, $J_{(2,3)} = 9.64$ Hz), 4.057 (1H, m, H_{rA}-2), 4.047 (3H, m, H_{rU}-3, H_{dA}-5, H_{rA}-5), 4.041 (2H, m, H_{rU}-2, H_{rA}-3, $J_{(1,2)} = 5.44$ Hz), 4.040 (1H, m, H_{dA}-6′, $J_{(6,6')} = 11.52$ Hz), 3.959 (1H, dd, H_U-4, $J_{(3,4)} = 9.44$ Hz, $J_{(4,5)} = 9.68$ Hz), 3.897 (1H, dd, $H_{\Delta U}$-2, $J_{(1,2)} = 8.48$ Hz, $J_{(2,3)} = 8.08$ Hz), 3.800 (1H, d, H_{rU}-1, $J_{(1,2)} = 5.44$ Hz), 3.756 (1H, dd, H_U-3, $J_{(2,3)} = 9.20$ Hz, $J_{(3,4)} = 9.44$ Hz), 3.695 (1H, d, H_U-5, $J_{(4,5)} = 9.68$ Hz), 3.665 (1H, dd, H_U-2, $J_{(1,2)} = 8.40$ Hz, $J_{(2,3)} = 9.20$ Hz), 2.060 (3H, s, H_{dA}-8), 2.057 (3H, s, H_{rA}-8), 1.384 (3H, d, H_F-6, $J_{(5,6)} = 6.48$ Hz), 1.328 (3H, d, H_{rF}-6, $J_{(5,6)} = 6.64$ Hz), 1.305 (3H, d, H_{dF}-6, $J_{(5,6)} = 6.56$ Hz); ^{13}C NMR (200 MHz, D_2O) δ 180.13 (C_{rU}-6), 177.83 (C_{dA}-7, C_{rA}-7), 177.81 (C_U-6), 171.72 ($C_{\Delta U}$-6), 149.94 ($C_{\Delta U}$-5), 109.20 ($C_{\Delta U}$-4), 106.74 ($C_{\Delta U}$-1), 106.00 (C_U-1), 103.60 (C_{rA}-1), 102.69 (C_{dA}-1), 101.35 (C_{rF}-1), 99.39 (C_F-1), 99.13 (C_{dF}-1), 84.00 (C_F-4), 83.90 (C_{rF}-4), 83.57 (C_{dF}-4), 83.14 (C_{rU}-4), 80.72 (C_{rU}-3), 80.06 (C_U-

3), 79.86 (C_U-5), 79.17 ($C_{\Delta U}$-3), 79.14 (C_{dA}-4, C_{rA}-4), 79.00 (C_{rA}-3), 78.86 (C_{dA}-3), 78.35 (C_U-4), 78.16 (C_{rF}-2), 77.91 (C_F-2), 77.85 (C_{dF}-2), 76.66 (C_U-2), 75.09 (C_{rU}-5, C_{dA}-5), 74.66 (C_{rA}-5), 73.47 (C_{rU}-2), 73.22 ($C_{\Delta U}$-2), 70.66 (C_{rA}-6), 70.22 (C_{dA}-6), 69.58 (C_{rF}-5), 69.41 (C_{rF}-3), 69.33 (C_{dF}-3, C_F-3), 69.25 (C_{dF}-5), 69.17 (C_F-5), 65.22 (C_{rU}-1), 54.39 (C_{dA}-2), 54.23 (C_{rA}-2), 25.44 (C_{dA}-8), 25,43 (C_{rA}-8), 18.83 (C_{rF}-6), 18.71 (C_F-6), 18.56 (C_{dF}-6); HRMS (ESI-Q-TOF) *m/z*: $[M - Na]^{2-}$ calcd for $C_{52}H_{69}N_2Na_{11}O_{70}S_{10}$ 1206.8991; Found 1206.8710.

4.9. Evaluation of anticoagulant activities in vitro

APTT, PT, and TT were determined with a coagulometer (TECO MC-4000, Germany) using APTT, PT and TT reagents and standard human plasma as previously described [7]. The compounds were dissolved in 20 mM Tris-HCl (pH 7.4) at various concentrations. For the APTT assay, 5 μL samples were mixed with 45 μL of normal human plasma, and incubated for 2 min at 37 °C. Then 50 μL of APTT reagent was added to the mixture, which was incubated for another 3 min at 37 °C. $CaCl_2$ (50 μL) was then added, and the clotting time was recorded. For the PT and TT assays, 5 μL samples (or 10 μL for TT) were mixed with 45 μL (or 90 μL for TT) of normal human plasma and incubated for 2 min at 37 °C; 100 μL of PT (or 50 μL of TT) reagent was then added and the clotting time was recorded.

4.10. Inhibition of human intrinsic factor Xase

Inhibition of intrinsic factor Xase (factor IXa–factor VIIIa complex) was determined by modified method with the reagents in the BIOPHEN FVIII:C Kit (HYPHEN BioMed) [3,7]. Different concentrations of glycosaminoglycan were incubated with 2 IU/mL factor VIII (30 μL), and factor IXa (30 μL) (containing human thrombin, calcium, and synthetic phospholipids) in a total volume of 90 μL, for 2 min at 37 °C. The reaction was initiated by the addition of factor X (30 μL) for 1 min at 37 °C. The amount of factor Xa generation was determined after 30 μL of factor Xa chromogenic substrate SXa-11 was added to the reaction mixture, based on the change of 405 nm optical density (OD). The OD_{405nm} was recorded every 30 s for 5 min at 37 °C using a Bio-Tek Microplate Reader (ELx 808, USA)). The change in absorbance/min (ΔOD/min) was then calculated. The blanks for kinetic measurement are also expressed as ΔOD/min and should give the highest values because they were carried out in the absence of active compound.

4.11. Inhibition of human thrombin and factor Xa by AT

Anti-thrombin and anti-factor Xa activities in the presence of AT were measured with Biophen Heparin Anti-IIa kits and Biophen Anti-Xa kits, respectively. A mixture of 30 μL samples and 30 μL 1 IU/mL AT was incubated at 37 °C for 2 min; 30 μL of 24 NIH/mL thrombin (or 8 μg/mL bovine factor Xa) was added. After incubation for 2 min (or 1 min for factor Xa), the residual thrombin or factor Xa activity was measured by addition of 30 μL of 1.25 mM thrombin chromogenic substrate CS-01(38) or 1.2 mM factor Xa chromogenic substrate SXa-11. The OD_{405nm} of the reaction mixture was recorded every 30 s for 5 min at 37 °C. The change in absorbance/min (ΔOD/min) was then calculated. The blanks for kinetic measurement are also expressed as ΔOD/min and should give the highest values because they were carried out in the absence of active compound.

Acknowledgements

We thank three anonymous reviewers for their helpful suggestions, which improved an earlier version of the manuscript. This work was funded in part by the National Natural Science Foundation of China (81673330, 81703374, and 81773737), Yunnan Provincial Science and Technology Department in China (2016FA050), Institutes for Drug Discovery and Development (CASIMM0220151008), and a grant from Youth Innovation Promotion Association (2017435), Discovery, Evaluation and Transformation of Active Natural Compounds, Strategic Biological Resources Service Network Programme (ZSTH-020) of Chinese Academy of Sciences.

Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.ejmech.2018.02.047.

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